

Some manufacturers of sensors for agriculture and environment monitoring

- AT&T
- Bosch
- DJI
- DROTEK
- ELISHA Systems Ltd
- Flux Farm
- Honeywell
- Infineon Technologies AG
- Phyttech
- PrecisionHawk
- Pycno
- Sentera
- Slantrange
- Smart Elements

can easily integrate onto drones. The sensor collects visual-band imagery, multispectral images as well as vegetation indices, including normalised difference vegetation index (NDVI) or normalised difference vegetation red edge (NDRE) data. It helps agronomists and farmers identify crop health issues with greater precision.

AI for farming

China and India are speeding up efforts in artificial intelligence (AI) innovation and development to assist farmers in planting decisions and crop protection. AI holds the promise of driving the agricultural revolution to produce more food using fewer resources. An AI system using Big Data management technologies and cloud computing aims to provide effective decision making for farmers.

Sensors monitor a fruit's progress in terms of perfect ripeness, adjusting light to accelerate or slow down the pace of maturation. This kind of farming requires considerable processing power. Intel's Xeon processors are being used to power AI algorithms.

Smart algorithms are applied to agricultural technologies to enable them to learn and teach themselves. AI and smart algorithms can analyse massive amounts of data on weather, environmental and historical information to make increasingly accurate predictions on what, where, when and how to plant crops for optimal performance and yields. There is great potential of AI combined with IoT4Ag.

Some smart agricultural sensors

Flux.IoT Eddy robot built with wireless sensors and image processing technology tells farmers specifics about their hydroponics and other indoor growing systems. It can monitor variables such as pH levels, temperature and relative humidity. It can also detect contaminants and provide information on how to resolve such problems.

Pycno's plug-and-play sensors with solar panels are fully-autonomous, having a simple dashboard. These allow seamless data collection and transfer from the farm to the smartphone.

Amber Agriculture smart sensor technology provides analytic solu-



Fig. 6: Double 4K sensor (Credit: <https://sentera.com>)

tions for the grain industry. Wireless pellet sensors distributed with the grain in storage bins or containers monitor such internal conditions as temperature, humidity and volatile compounds.

Acuity Agriculture wireless field sensors allow farmers to monitor soil conditions and pest development, schedule irrigation events, track climate conditions and predict harvest timings.

Phyttech smart sensors placed directly on plants and crops can monitor the micro-variations of stem diameters to provide stress indications.

Slantrange sensor and analytic systems are used on drone aircraft to measure crop and weed densities, stress from pest infestations, nutrient deficiencies and dehydration levels.

Environmental monitoring sensors

Climate and weather are the top influencing factors in the production of crops. Sensors are getting ubiquitous and playing an important role in sensing the environment. These include sensors for monitoring water, air, weather and soil.

Barometric sensors are used in weather networks. Infineon Technologies manufactures barometric pressure sensors for mobile and wearable devices having high precision and low power consumption. For example, their DPS310 barometric pressure sensor is recommended for applications where power consumption is critical and highest precision in pressure metering is required.

With an IoT-enabled weather monitoring system, farmers can determine the optimum time for plantation, irrigation and harvesting. Predictive weather models are constructed using sensors in IoT applications. Based on sensor data, farmers can plan and change harvest and irrigation times to improve the crops. By organising and analysing collected data, they can take pre-emptive actions for healthy harvest of their crops.

To sum up

We are entering the era of Agriculture 4.0, which is a new phase of the agriculture revolution. We are witnessing precision, smart and digital farming using various modern technologies. Recent technology trends include sensors for connected livestock, satellites, the IoT and GPS sensors, among others, to measure a variety of essential soil and plant properties on-the-go.

Scanning from the sky using drones attached with camera sensors to monitor the fields, and also monitoring from anywhere using the Internet and IoT sensors, farmers can access various information related to real-time sensing of soil and crops.

AI with modern sensors can be used to adapt strategies to the changing conditions of the field and environment. Sensors empower farmers to react quickly and dynamically maximise crop performance, resulting in better yields and higher food productivity. **EFY**